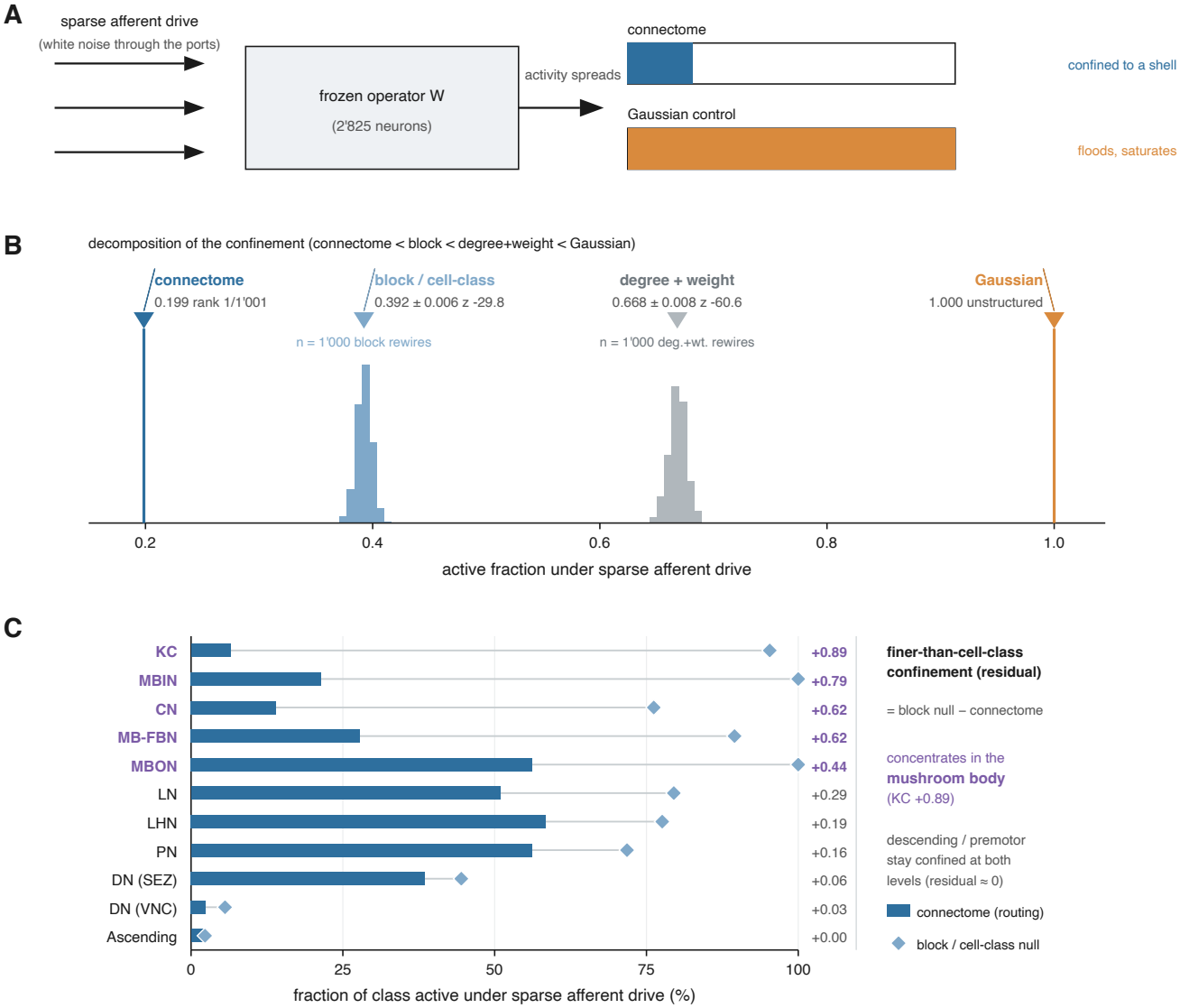

A
sparse afferent drive
(white noise through the ports)
frozen operator W
(2'825 neurons)
activity spreads
connectome
Gaussian control
confined to a shell
floods, saturates
B
decomposition of the confinement (connectome < block < degree+weight < Gaussian)
connectome
0.199 rank 1/1'001
block / cell-class
0.392 ± 0.006 z -29.8
n = 1'000 block rewires
degree + weight
0.668 ± 0.008 z -60.6
n = 1'000 deg.+wt. rewires
Gaussian
1.000 unstructured
0.2
0.4
0.6
0.8
1.0
active fraction under sparse afferent drive
C
KC
MBIN
CN
MB-FBN
MBON
LN
LHN
PN
DN (SEZ)
DN (VNC)
Ascending
+0.89
+0.79
+0.62
+0.62
+0.44
+0.29
+0.19
+0.16
+0.06
+0.03
+0.00
0
25
50
75
100
fraction of class active under sparse afferent drive (%)
finer-than-cell-class confinement (residual)
= block null – connectome
concentrates in the mushroom body (KC +0.89)
descending / premotor stay confined at both levels (residual ≈ 0)
connectome (routing)
block / cell-class null